

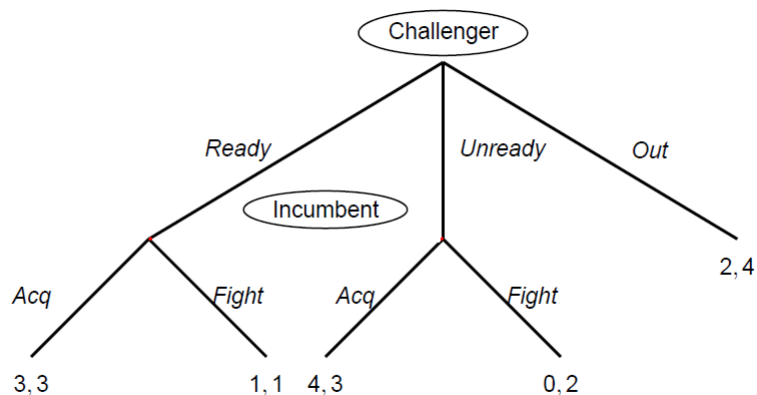
Problem Set 3

POLS W4732

Fall 2026

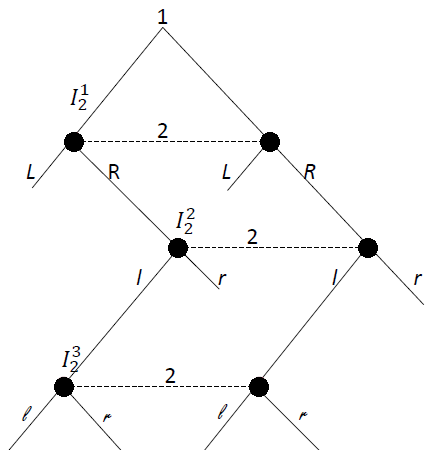
Due October 16 before the TA section

Exercise 1. Consider the following version of the Entry Game



Using its strategic form representative, find the set of Nash Equilibria of this game.

Exercise 2. Consider the following game, where I_2^1, I_2^2, I_2^3 denote the information sets for Pl. 2. Let $s_2 \equiv (L, l, r)$ and $s'_2 \equiv (R, r, l)$, denote two pure strategies for Pl. 2. Find the behavioral strategy equivalent to the mixed strategy that puts equal probability (i.e., $1/2$ and $1/2$) on s_2 and s'_2



Exercise 3. Consider the Public Good Provision Game, under the assumption that both costs are drawn from a uniform distribution in $[\gamma - \frac{1}{2\psi}, \gamma + \frac{1}{2\psi}]$, with $\gamma - (2\psi)^{-1} < 1 < \gamma + (2\psi)^{-1}$

1. Characterize the unique Nash Equilibrium
2. How does the equilibrium threshold c^* change with ψ and γ ? Can you explain these results intuitively?
3. How does the equilibrium probability that the public good is provided change with ψ and γ ? Can you briefly explain these results intuitively?
4. Now suppose that instead of two players, the game features n players. For simplicity, also assume that $\psi = 1/2$ and $\gamma = 1$. How does the equilibrium threshold c^* change with n ? Can you briefly articulate an intuition for this result?
5. (**hard**) Under the same assumptions as in the previous point, how does the probability that the public good is provided change with n (provided n is arbitrarily large)? Can you briefly articulate an intuition for this result (even if you are unable to show this)?

Exercise 4. Consider the following simple model of war. There are two countries (1 and 2). Each has two actions: F (prepare for a fight) or N (accommodate). We define war as the outcome (F, F) . When both countries choose N , they obtain a payoff of 0 for sure. Choosing F incurs mobilization cost μ whether or not the war happens. If the war happens, the winner takes resources worth 1 from the loser. The winner is ex ante uncertain, and is the realization of a state of nature $\omega \in \{UR, DR, UL, DL\}$. When the state is either UR or DL , country 1 would win the war should it occur; otherwise, country 2 would win. We assume that each country only observes one of the two components of the state (e.g., when the state is UL country one receives signal “U”, country 2 receives signal “L”).

$$\begin{aligned} u_i(N, a_j, \omega) &= 0 \quad \forall (a_j, \omega) \in \{F, N\} \times \Omega \\ u_i(F, N, \omega) &= -\mu \quad \forall \omega \in \Omega \\ u_i(F, F, \omega) &\in \{1 - \mu, -1 - \mu\} \quad \forall \omega \in \Omega \end{aligned}$$

Notice that fighting is inefficient: $u_1(F, F, \omega) + u_2(F, F, \omega) = -2\mu$ for all $\omega \in \Omega$.

1. Show that $(N(U), N(D), N(L), N(R))$ is a Bayesian equilibrium
2. Show that, if country 1 chooses F when its signal is U , it must be that the other country also chooses F with positive probability (either upon seeing R or upon seeing L , or both)
3. Show that if this is the case, at least one country has a profitable deviation